

Answer **all** questions

Write your answers in the spaces provided.

- 1** Malaria is a disease which is spread by mosquitoes. Malaria is caused by a single-celled organism. In an investigation, a gene from a bee was transferred into mosquitoes, using a plasmid. This gene prevented the mosquitoes and their offspring from spreading malaria.

(a) Describe how the bee gene could have been inserted into the plasmid.

[3]

(b) These plasmids were injected into the fertilised eggs of mosquitoes, rather than into adult mosquitoes. Explain why.

[2]

(c) Within the plasmid contained a genetic marker, a gene encoding a fluorescent protein that produces green light. Explain why this marker gene was used in this experiment.

[2]

(d) Many years ago humans accidentally introduced a species of mosquito to some Hawaiian islands. This species of mosquito carries a type of malaria that only infects birds. Some of these birds are rare. It has been suggested that the control of malaria using genetically-engineered mosquitoes with the bee gene should be tested on these Hawaiian islands. Suggest **one** advantages of using this approach.

[1]

The polymerase chain reaction (PCR) is used to make many copies of a DNA. The steps in each cycle of the reaction take place at different temperatures.

(e) Explain why during each cycle,

(i) the DNA is heated to 95 °C.

_____ [1]

(ii) then reaction mixture is cooled to 40 °C.

_____ [1]

(iii) and the temperature is again raised, but to 70 °C.

_____ [1]

(f) In polymerase chain reaction, suggest why DNA synthesis takes place in opposite directions on the two separated strands of the double helix.

_____ [2]

(g) State **one** advantage and **one** disadvantage of using *Taq* DNA polymerase.

_____ [2]

[Total: 15]

- 2 In an investigation, DNA was extracted from the remains of a soldier who had died in a war. His body was blown beyond recognition and thus restriction fragment length polymorphism (RFLP) analysis was done to determine his identity.

DNA samples of four different couples with missing sons from the war were isolated and analysed to determine his real identity. The banding patterns produced on the gel for analysis is as shown in **Fig. 2.1**

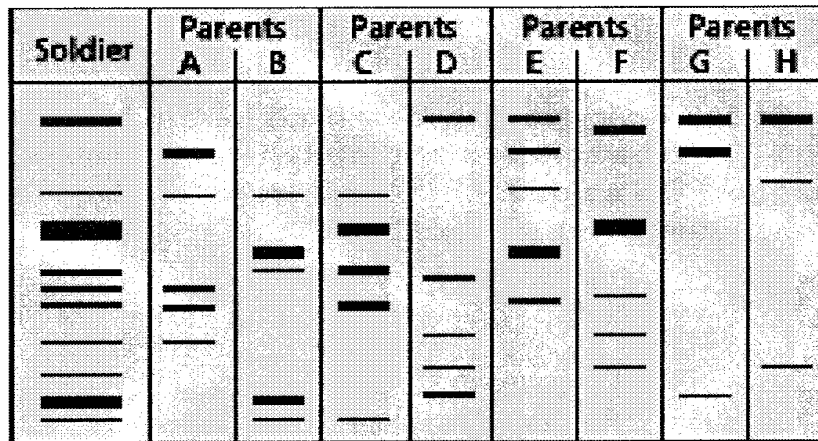


Fig. 2.1

(a) Explain briefly:

- (i) How the DNA fragments are separated by gel electrophoresis.

[2]

- (ii) The basis behind RFLP.

[2]

- (b) A marker DNA should be loaded onto the gel together with these DNA samples. State the role of a marker DNA.

[1]

(c) A southern blot was prepared and probes, specific for certain RFLP markers in human beings, were added.

(i) How do probes detect the specific RFLP markers present on the different individuals?

[2]

(ii) Give an example of an RFLP marker used in forensic science or paternity testing.

[1]

(d) Which pair of parents is likely to be the parents of the dead soldier? Explain your choice.

[2]

A length of DNA from one of a pair of homologous chromosomes from a dog is shown in **Fig. 2.2**. The target sites of *Eco*RI are shown by arrows and the length of DNA between target sites given kilobases (kb).



Fig. 2.2

A mutation alters one base of the restriction site marked with an asterisk (*). DNA from one of the dogs which is heterozygous for this mutation is cut with *Eco*RI and the DNA fragments separated by gel electrophoresis.

(e) State

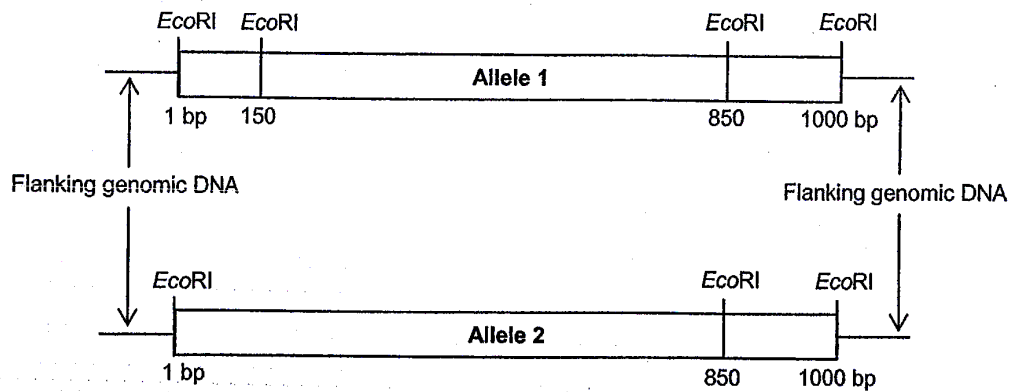
- (i) the number of bands, seen on the gel, from this length of DNA produced from the pair of homologous chromosomes.

_____ [1]

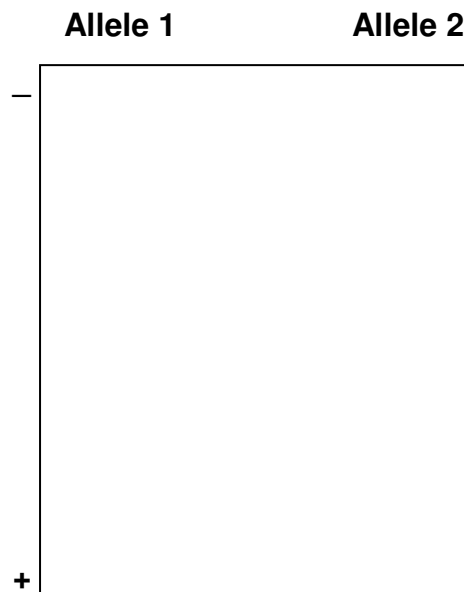
- (ii) The different sizes of these fragments.

_____ [1]

- (f) The diagram below shows the restriction site present in two alleles that are 1000bp long.



- (i) On **Allele 1**, indicate the position of a probe that you would design to distinguish the two alleles. [1]
- (ii) In the gel below, draw the resulting autoradiogram you would expect to see based on the probe you have drawn. [2]



[Total: 15]

- 3** A variety of vectors have been developed and used in gene therapy to correct defective genes / alleles, most notably in somatic cell gene therapy. Such vectors include viral and non-viral vectors.

Some of the most successful cases of genetic therapy involve the use of retroviruses in the treatment of X-linked severe combined immunodeficiency (X-linked SCID). Although the treatment is highly successful on the whole, leukaemia develops subsequently in a significant number of the patients.

Adenoviruses, on the other hand, have been used to treat cancer. A distinguishing feature of using adenovirus is that the gene being transferred into a cancer cell is not incorporated into its genome.

Non-viral treatments include the use of naked DNA and liposomes, made up of lipids in a manner similar to the plasma membrane. A certain degree of success has been achieved in gene therapy using liposomes as vectors to transfer genes to cancer and respiratory epithelial cells.

- (a)** Describe what is meant by somatic cell gene therapy.

_____ [1]

- (b)** Explain what is meant by an X-linked disease, and suggest how such diseases are first noticed in populations.

_____ [2]

- (c)** Describe two properties of retroviruses that make them suitable as vectors for gene therapy.

_____ [2]

- (d)** Explain why patients treated with retroviruses are at significant risks of contracting leukaemia.

_____ [2]

- (e) Explain how DNA transferred by adenoviruses could help patients even though the DNA is not incorporated into the genome.

[2]

- (f) Elaborate on a feature of the liposome that makes it suitable as a vector for gene transfer.

[2]

- (g) Based on the information provided, describe how liposomes could be used to treat a named respiratory disease.

[3]

- (h) Based on the understanding that every current vector for gene therapy has its strengths and weaknesses, suggest how future vectors for gene therapy might be like.

[1]

[Total: 15]

Write your answers to this question on the separate answer paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 (a)** Describe how genetic engineering has helped in improving the quality and yield of named crop plants and animals to solve the demand for food (in terms of quantity and quality) in the world. [8]
- (b)** Discuss the potential risks of genetically modified organisms. [6]
- (c)** Explain the normal functions of a named type of adult stem cell. [6]